

ReRAM: History, Status, and Future

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Abstract—This article reviews the resistive random-access memory (ReRAM) technology initialization back in the 1960s and its heavily focused research and development from the early 2000s. This review goes through various oxygen/oxygen vacancy and metal-ion-based ReRAM devices and their operation mechanisms. This review also benchmarks the performance of various oxygen/oxygen vacancy and metal-ion-based ReRAM devices with general trend drawn. Being a semiconductor memory and storage technology, the commercialization attempts for both stand-alone mass storage/storage-class memory and embedded nonvolatile memory are also reviewed. Looking toward the coming era, the potential of using ReRAM technology to improve machine learning efficiency is discussed.

Index Terms—Conductive bridge ReRAM (CBRAM), filament, machine learning, NAND, nonvolatile memory (NVM), OxRAM, ReRAM, storage-class memory (SCM), self-rectify cell, storage.

I. INTRODUCTION

RESISTIVE random-access memory (ReRAM), as a resistive switching memory, covers a broad range of memory and storage types of semiconductor devices. Generally speaking, resistive switching memory includes any devices with resistance change under external stress. For the technology that is reviewed in this article, the focus is on the insulator-based material stacks with defect-based nature as switching species. Fig. 1 highlights the review focus among the various resistive memory devices. The defects used as switching media are either generated from the insulating layer internally (i.e., oxygen vacancy in oxide dielectrics by electrical soft breakdown) or introduced from the electrode layer externally (i.e., Cu ions into electrolytes by electrical stress).

Using defects in a memory device is not unusual. For example, 3-D NAND flash memory uses SiN as a charge trap layer for electrons. However, moving defects back and forth, as in ReRAM, is still unique. The defect-based nature brings ReRAM lots of unique properties for memory application.

II. HISTORY OF ReRAM

A. Back in the 1960s

Studies of the ReRAM device on various oxide materials, i.e., Al_2O_3 , NiO , SiO_2 , Ta_2O_5 , ZrO_2 , TiO_2 , and Nb_2O_5 , were

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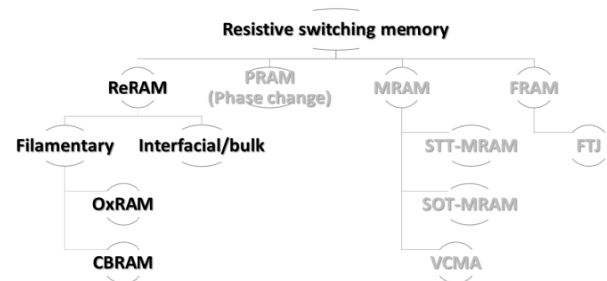


Fig. 1. General category of resistive switching memory technologies with ReRAM highlighted as the review focus.

reported back in the early 1960s [1]–[4]. With that said, the history of the ReRAM device is as long as that of the metal–oxide–semiconductor field-effect transistor (MOSFET) (1960) [5], [6], and longer than both dynamic random access memory (DRAM) (1966) [7] and floating gate (FG) memory (1967) [8]. However, the research on the resistive switching phenomena in a dielectric material did not point to a clear application at that time. More importantly, the mainstream storage technology from the 1950s till the early 2000s is either magnetic based (tapes, hard drives, etc.) or light based (CD, DVD, etc.). Therefore, the study of ionic switching in a solid-state device is largely confined to academic exploration.

B. From the 2000s

The explosion of portable electronic devices, particularly smartphones, drives the demand of NAND flash memory as storage media from the mid-2000s. With the aggressive scaling of 2-D NAND flash memory [9], what would be the technology to extend the semiconductor storage scaling toward even higher density, even faster speed, and even lower cost was put onto the discussion among industry and academy. ReRAM as one of the proposals to continue the scaling trend of NAND flash memory as mass storage has started to attract attention again. Though over the past 20 years of development, the application of ReRAM is diversified into several other domains, i.e., embedded nonvolatile memory (eNVM) and storage class memory (SCM), the original intention of ReRAM research and development is meant for NAND flash extension. Section VI will discuss this point further.

The decade between 2005 and 2015 was the golden era of ReRAM research and development. Fig. 2 lists the major milestones of ReRAM development: the first NiO_x -based ReRAM with promising device characteristics and reliability [10], the first integrated conductive bridge ReRAM (CBRAM) [12], the first HfO_2/Ti device with fully conventional fab materials [13], the first published proposal

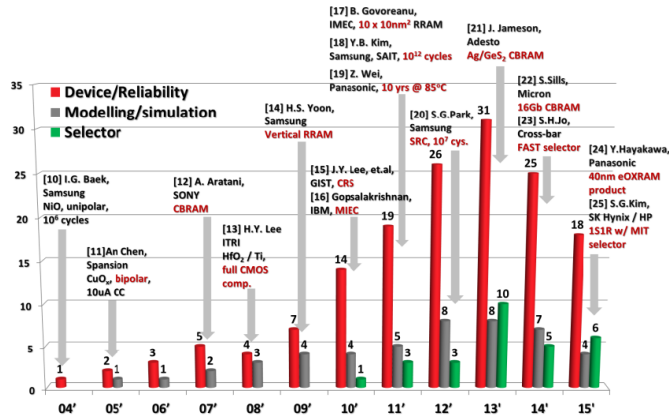


Fig. 2. All ReRAM technology publications in IEDM, VLSI (VLSI Technology Symposium) and [Nonvolatile Semiconductor Memory Workshop/IMW (NVSMW/IMW)] between 2004 and 2015, with major demonstration milestones highlighted [10]–[25].

of 3-D vertical ReRAM [14], the first scaled $10\text{ nm} \times 10\text{ nm}$ ReRAM integrated in 12-in fab [17], the first 16-Gb CBRAM integrated chip [22], the first 40-nm node embedded ReRAM product [24], etc.

C. Future of NAND Is NAND. What Can ReRAM Do?

With the aggressive logic-state scaling from a single-level cell (SLC) to a multiple-level cell (MLC) to a triple-level cell (TLC) in 2-D NAND flash memory [9], as well as even more aggressive physical scaling down to 15-nm critical dimension (CD), the insertion point of ReRAM on the storage roadmap keeps slipping away. Moreover, the introduction [26] and commercialization [27] of 3-D NAND technology pave a path for NAND scaling in the coming decade and further raises the entry requirement of ReRAM. The ReRAM society starts to diversify the application into the low density and low-cost eNVM domain to partially compete with NOR flash and create more Internet of Things (IoT) device adoption. The concept of SCM gives another hope to ReRAM in an intermediate density range target (i.e., 32–128 Gb per die). However, the SCM target sees challenges from both memory cell and selector integration in a high-density architecture. The eNVM application has been pursued by several affiliations [24], [28]–[30], but its adoption is rather gradual. After 2015, ReRAM technology research and development start to gradually wave down.

D. If Not Memory, Anything Else?

With the revival of machine learning in the 2010s, ReRAM for either a binary or an analog computing/storage unit cell has been heavily studied over the past five years. The potential of ReRAM in machine learning and computing in memory (CiM) is there. But its commercial adoption is still at the path-finding phase. Section VII will briefly discuss ReRAM in machine learning application.

III. ReRAM DEVICE PHYSICS

A. Device Physics: Filamentary Switching

As shown in Fig. 1, there are two types of filamentary ReRAM devices: OxRAM (Oxide ReRAM) and CBRAM.

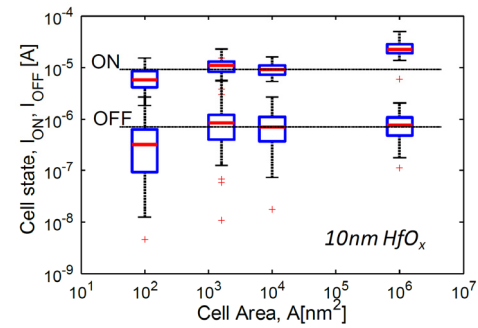


Fig. 3. ReRAM cell switching current (HfO_2 based OxRAM) versus cell physical size (from $1\text{ }\mu\text{m} \times 1\text{ }\mu\text{m}$ down to $10\text{ nm} \times 10\text{ nm}$) [17].

Both of them typically show switching current independence of device size. This is a highly desired property of a memory device with physical scaling demand. From the early work of the integrated transition metal oxide (TMO)-based ReRAM [10] to the $10\text{ nm} \times 10\text{ nm}$ extremely scaled cell [17], the switching current independence of the cell size has been demonstrated. Once the switching current limit is determined by the load resistance, selector transistor, or other external parts, the ReRAM switches at this predetermined current level without clear physical cell-size-scaling impact.

Based on this simple electrical characteristic, a localized conductive filament is proposed to be the current conduction path inside the ReRAM cell. The filament size is imaged to be smaller than $10\text{ nm} \times 10\text{ nm}$, to explain the scaling independence demonstrated down to that dimension (Fig. 3).

The filamentary switching theory is well accepted among the industry and academia. The modeling efforts [31]–[36] are mostly spent on the filament creation and switching processes (Table I summarizes the different modeling efforts in [31]–[36]). The engineering efforts are mostly spent on how to better control the filament creation and switching to improve the uniformity and stability.

With the introduction to filament theory, a detailed discussion on its creation and switching processes is followed.

1) Filament Forming: To achieve a revisable switching, the filament is created in a controlled manner. A current-limiting mechanism is typically applied during its first creation (referred to as FORMING) and consecutive recreations (referred to as SET). Since ReRAM is usually made from insulating materials, i.e., oxides, the FORMING and SET processes are similar as a controlled dielectric soft breakdown. The current-limiting mechanism is typically a current compliance applied from the external characterization instruments, or load resistance/selector transistors on-chip integrated, or off-chip connected with the ReRAM cell. As the filament is created during the very first FORMING step, filament properties are intuitively thought to be strongly dependent on this process.

A few questions that physicists typically raise are as follows:

- 1) What is the physical species created after FORMING?
- 2) Can we see them physically?

A few questions that the engineers typically raise:

- 1) How high is defect density within the filament?
- 2) How to make a filament with high defect density?

TABLE I
SEVERAL REPRESENTATIVE PHYSICAL MODELS OF EXPLAINING AND SIMULATING THE OxRAM SWITCHING PROCESS [31]–[36]

Models	Peking Univ. [31]	Poly. Tech. Milano [32]	Stanford Univ. [33]	SEMATECH [34]	DISMI [35]	Imec [36]
Approach	Phenomenological / Monte-Carlo	Phenomenological / Analytical	Phenomenological / Monte-Carlo	Phenomenological	Ab-initio / Phenomenological	Ab-initio / Phenomenological
Medium	V_o / O^{2-}	Thermal heating, V_o / O^{2-}	V_o / O^{2-}	V_o / O^{2-}	V_o / O^{2-}	V_o
SW mechanism	Redox / depletion-recomb.	Joule-heating	Redox	Redox	Redox	V_o drift
Bipolar / Unipolar	Bipolar / Unipolar	Unipolar / Bipolar	Bipolar	Bipolar	Bipolar	Bipolar

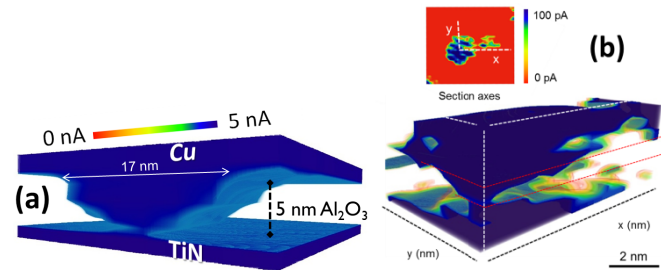


Fig. 4. 3-D visualization of filament shape with advanced c-AFM technology. (a) Cu filament in CBRAM [37]. (b) Oxygen vacancy filament in OxRAM [38].

Physicists' questions are based on the nature of defects consisting the filament. As the FORMING is a controlled dielectric soft breakdown process, the nature of the species created is commonly accepted as oxygen vacancies (in OxRAM case) and metal ions from the electrode (Cu, Ag, etc., in the CBRAM case). There are seldom other proposals on the defect nature, but rather detailed discussion on (in the case of OxRAM) whether it is an oxygen atom moving out of the oxide lattice (substitutional oxygen vacancy) or a metal atom moving into the oxide (interstitial oxygen vacancy).

The efforts in visualizing the filament have been spent with different techniques used. The most direct visualization in 3-D is based on the conductive atomic force microscopy (c-AFM) analysis [37], [38], where sizes in both the CBRAM and OxRAM filaments are observed in a single-digit nm range (Fig. 4).

In terms of engineering methods to better control the filament properties during FORMING, a few different FORMING processes were proposed to increase the defect density within the filament. Though ReRAM does not have the same lateral scaling limitation as does 2-D NAND flash, the switching current reduction for a high-density memory architecture sets the scaling challenge for ReRAM. The filament stability depends on the total number of defects consisting the filament, and scales with the maximum FORMING and switching current. With bit density scaling-up requirement for ReRAM, the switching current itself needs to be scaled down, preferably below 10 μ A for mass storage or SCM. Defects-based filament shows increasing nonuniformity and much shorter retention when the number of defects is scaled [39], [40]. To counter engineer this switching current issue, FORMING with multiple step [41], reverse bias [41], at higher temperatures [42], etc. were proposed. Improvement was seen for typical TMO stacks.

Another engineering topic related to the FORMING process is the "FORMING-free" device. Typically, the FORMING process requires a higher voltage and energy than the

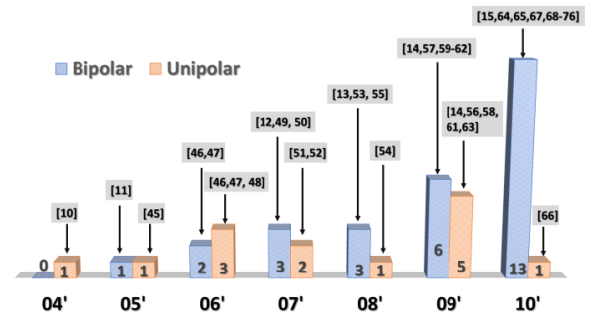


Fig. 5. Statistics of bipolar versus unipolar switching ReRAM device demonstration from IEDM, VLSI, NVSMW/IMW between 2004 and 2010.

SET process. This brings practical limitation to integrating ReRAM with transistors or selectors, as the FORMING voltage can be too high for them to operate properly. To free this limit, numbers of engineering efforts were spent on bringing the FORMING voltage down to the similar range of the SET voltage. By thinning down the dielectric layer or create more defects during dielectric fabrication, the FORMING voltage can decrease. The penalty of using a thinner dielectric layer or a higher defect density dielectric layer is that the resistance ration between the low resistance state (LRS) and high resistance state (HRS) becomes smaller [43]. Physically, a forming free stack has already certain number of leakage paths during device fabrication, and background leakage from the thin dielectric layer limits the HRS.

2) Filament Switching: After filament is created, the cells are subject to reversible switching between LRS and HRS by SET (HRS to LRS) and RESET (LRS to HRS) operations. Filamentary switching can either be unipolar (SET and RESET in the same voltage polarity) or bipolar (SET and RESET in the opposite voltage polarity).

In early years, unipolar switching, by leveraging the physical model and characterization know-how using the phase change memory technology, was popular among devices (Fig. 5). Unipolar switching is generally regarded as the thermal heating assistant mechanism for its RESET process [44]. As the RESET process requires Joule heating to melt down the filament, switching current and current density are challenging to be scaled down. The engineering focus on ReRAM gradually shifted to bipolar switching after 2010 (Fig. 5).

It is worth mentioning that, besides the Joule-heating-based mechanism, electron trapping and detrapping was also considered for unipolar switching [10], [11]. However, the electron trapping and detrapping process is more associated with dielectric to electrode Schottky barrier height modification. It can coexist with the filament present in the device but does

TABLE II
CONDUCTION MECHANISM PROPOSED FOR ReRAM LRS AND HRS STATES

Mechanism	Ohmic	Space Charge Limited current (SCLC)	Poole-Frenkel	Schottky emission	Trap-assistant tunneling	Quantum point contact	Fowler-Nordheim / direct tunneling
Current range	Large	Large	Large - medium	Large - medium	Medium - low	Medium - low	Low
Area depend.	Y	Y	Y	Y	N	N	Y
Temp. depend.	$e^{(-c/T)}$	N	$e^{(-A/T)}$	$T^2 e^{(-A/T)}$			
E-field depend.	V	V^2	$V^* e^{(V^{0.5})}$	$V^* e^{(V^{0.5})}$	$V^2 e^{(-A/V)}$	$e^{(A*V)}$	$V^2 e^{(-A/V)}$

not fully explain the relationship between the cell size and switching current (weakly dependent or independent).

Bipolar switching in most of the oxide-based material systems is based on the reduction–oxidation process (referred to as redox) via ion movement. Fast-movable ions are either oxygen vacancies in OxRAM or highly diffusive metal ions in CBRAM (Cu, Ag, Co, etc.). Thermodynamically, the redox-based switching mechanism defines the SET and RESET operations in exchanging the complementary types of species (i.e., oxygen versus oxygen vacancy) in a complementary switching polarity. Thus, the redox-based bipolar switching is regarded more sustainable over SET/RESET operations. Numbers of long endurance ($>10^{10}$ cycles) were reported on OxRAM devices [18], [69], [77], [78], which points to this advantage.

Studies on SET and RESET mechanisms are closely linked to the engineering work on operation voltage determination for array design as well as endurance improvement.

For array design, the SET and RESET voltages are preferred to be stable, particularly at different temperatures and different endurance cycles. From the understood mechanism, both the redox reaction and ion transportation are thermally dependent, though experimentally the temperature dependence on SET and RESET voltages is marginal for ReRAM. It is probably because the redox and ion transportation within the nm range take place at such faster speeds that no clear impact is visible on the SET and RESET.

For endurance improvement, the SET and RESET operations in bipolar switching lead to the general understanding of balancing the number of ions transported back and forth. As the redox reaction and forth back ion transportation are both complementary, balancing the SET and RESET operation with tuning the switching voltage or switching time or switching current can improve the endurance [77]. The concept of balancing SET and RESET operations becomes a standard engineering work for ReRAM endurance improvement.

3) Current Conduction of LRS and HRS: Current conduction mechanisms for LRS and HRS are counted for READ operation and the LRS and HRS distributions. It is somewhat overlooked with respect to SET and RESET switching, as it is a static physical event. From the various fit and modeled mechanisms for LRS and HRS (Table II), one general conclusion is that the LRS and HRS conduction mechanisms are probably dependent on the current. Take LRS as an example. For large current operation ($>200 \mu\text{A}$), the LRS conduction is mostly ohmic. A filament with a high number of defects is created with a high switching current, and the total defect number consisting of the filament is large. The distribution

of LRS is very tight and the retention is stable. With the switching current scaling (medium current $50\text{--}200 \mu\text{A}$ and low current $<50 \mu\text{A}$), the LRS conducting current is decreasing. The filament thus requires fewer defects to conduct the lower current. In part of the filament, not all defects are well overlapped with each other. The conducting current shows more nonlinear behavior. Nonlinear I – V itself is not an issue but fewer defects in the filament are an issue. With fewer defects, the distribution of LRS without verifications becomes broader, and the retention starts to seriously degrade.

B. Device Physics: Interfacial/Bulk Switching

As illustrated in Fig. 1, besides the filamentary switching devices, there are also ionic switching devices that show the dependence of the switching current on the cell size. The switching mechanism is understood as the dielectric to electrode Schottky barrier height modulation or redox at dielectric bulk with the conductance change of the stack. The interfacial/bulk switching is typically bipolar. As shown in Fig. 2, area-dependent type of switching is always present in technology demonstration, but it always stays under the shadow of filamentary switching. Owing to the requirement of moving more ions for the interfacial/bulk switching mechanism, area switching is usually slower than filamentary switching. However, with the switching current reduction of filamentary ReRAM becoming more difficult below $50 \mu\text{A}$, interests on interfacial/bulk switching have risen over the past five years.

Besides the different dependence of the switching current on the cell size, interfacial/bulk switching also has a smaller switching current and more stable retention at a low current range [79], [80]. Moreover, the interfacial/bulk switching cell typically shows higher nonlinearity than the filamentary cell, which provides certain self-rectifying features in array operation. These advantages make interfacial/bulk switching more suitable for large-density and low-current operation memory array. Section V will provide a more detailed technology benchmark on filamentary versus area switching.

IV. ReRAM MATERIALS

A. Materials of the Switching Layer

Various materials have demonstrated resistive switching phenomenon. Table III summarizes the elements used in their oxide form as the main switching layer. Since one of the dominant switching mechanisms is redox-based filamentary switching, oxides and particularly TMOs are the most common materials. Simply put, if only oxygen vacancies are needed

TABLE III

ReRAM DIELECTRIC AND ELECTRODE MATERIALS DEMONSTRATED IN LITERATURES, WITH SWITCHING MODES LISTED

Dielectrics	Switching polarity	Electrode Materials	Switching polarity
NiO	Unipolar ^[81] / Bipolar ^[82]	Pt	Unipolar ^[89] / Bipolar ^[89]
TiO ₂	Unipolar ^[83] / Bipolar ^[84]	Ag	Bipolar ^[111]
Nb ₂ O ₅	Unipolar ^[85] / Bipolar ^[86]	Au	Unipolar ^[112]
ZrO ₂	Unipolar ^[87] / Bipolar ^[88]	Ir	Bipolar ^[53]
HfO ₂	Unipolar ^[89] / Bipolar ^[17]	Cu	Bipolar ^[113]
Ta ₂ O ₅	Unipolar ^[90] / Bipolar ^[53]	Co	Bipolar ^[114]
MgO	Unipolar ^[91] / Bipolar ^[92]	Ru	Unipolar ^[115]
Al ₂ O ₃	Unipolar ^[93] / Bipolar ^[94]	Ni	Unipolar ^[116] / Bipolar ^[117]
CoO	Unipolar ^[95] / Bipolar ^[76]	Ti	Bipolar ^[13]
ZnO	Unipolar ^[96] / Bipolar ^[97]	Zr	Bipolar ^[15]
SnO ₂	Unipolar ^[98] / Bipolar ^[99]	Hf	Bipolar ^[17]
CuO	Unipolar ^[100] / Bipolar ^[111]	Ta	Bipolar ^[118]
FeO	Bipolar ^[50]	TiN	Bipolar ^[119]
CeO	Unipolar ^[101] / Bipolar ^[59]	Al	Bipolar ^[120]
WO	Unipolar ^[102] / Bipolar ^[49]	La	Unipolar ^[121]
GeO	Unipolar ^[103] / Bipolar ^[104]	W	Bipolar ^[49]
STO	Unipolar ^[73] / Bipolar ^[105]	CNT	Bipolar ^[122]
SiO ₂	Unipolar ^[106] / Bipolar ^[107]	Graphene	Bipolar ^[133]
GdO	Bipolar ^[78]		
BN	Bipolar ^[108]		
AlN	Bipolar ^[109]		
ZnSe:Ge	Unipolar/Bipolar ^[110]		

for switching, there is no major difference between different oxide materials. The cousins of oxides, nitrides [108], [109], and some of the chalcogenides [110] also demonstrate resistive switching with nitrogen and Te/Se/S vacancies.

Materials are directly linked to the integration work on the wafer level in the semiconductor fabs. As introduced in Section II, the revival of ReRAM in the 2000s was mainly driven for one clear application as NAND flash extension. Therefore, the material stacks need to be feasible and easy for integration on the wafer level. Even though, through tremendous processes, the technology progress was made, the number of materials typically dealt with in a 12-in fab is still limited. Oxide materials present in the fab are predominantly SiO₂. Thanks to the logic Complementary Metal Oxide Semiconductor (CMOS) scaling, high-*k* oxides such as HfO₂ are introduced into the fab. Though HfO₂ is still has a higher cross-contamination risk than SiO₂, the fabs have know-how to integrate it. HfO₂, thus, becomes a popular choice as a ReRAM material stack for the integrated device and array.

For the same oxide material, i.e., HfO₂, there are various ways of depositing the materials and integrating them under different thermal budgets. Different process conditions for the same material change the properties of the material. A reactive physical vapor deposition sputtered HfO₂ has flexibility in tuning the Hf:O ratio and modify the defect density during the reaction process. An atomic layer deposition HfO₂ is close to stoichiometry, offering excellent film uniformity and coverage in a 3-D vertical structure, but it requires an extra scavenging layer to create certain defect density for switching.

With the ReRAM development going on, the technology direction gradually splits into stand-alone and embedded applications. In both cases, the ReRAM integrates with the back end of line (BEOL) process (in the definition of logic CMOS technology). The ReRAM layer is sitting a few BEOL insulator and metal layers above the CMOS selector transistor and periphery transistor layer. Thanks to this isolation, the contamination risk of ReRAM stack materials to the front end of line (FEOL) CMOS is relatively low.

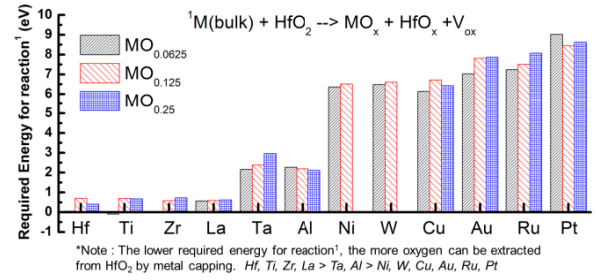


Fig. 6. *Ab initio* simulation of required energy for reduction reaction (1). The lower the energy, the easier the reaction to happen [118].

B. Materials of the Electrodes

Electrodes in the ReRAM stack are not only for electrical contact, but also are an important part of the resistive switching layer. The understanding of the role of active electrodes in the ReRAM stack remained a focus area during the early development stage of this technology. The demonstrated electrode materials are listed in Table III. They can be divided into three groups:

- 1) Noble metal, i.e., Pt, Au, Ir, etc.
- 2) Active metal for OxRAM, i.e., Ti, Zr, Hf, Ta, Al, Ni, etc.
- 3) Active metal for CBRAM, i.e., Cu, Ag, Co, etc.

Take OxRAM as an example to explain the role of the electrodes. Early demonstrations of a ReRAM device show that it majorly consists of noble metal electrodes. The SET operation of the redox switching mechanism creates oxygen vacancies and pushes the same amount of oxygen atoms pushed out of their original lattice position toward the anode. These oxygen atoms are needed during RESET operation to oxidize the oxygen vacancies. Thus, when they reach the anode, they stay as separate atoms rather than bonding to the electrode materials. Noble metals do not react with oxygen, and thus suitable as electrodes to preserve oxygen atoms for reservable switching. They offer stable resistive switching owing to their inner chemical properties. However, they are all difficult to be integrated in a conventional CMOS process.

C. Switching Layer and Active Electrodes' Interaction

When active electrodes are in contact with the switching layer, i.e., Ti to HfO₂, active electrode materials are able to take certain amount of oxygen from the HfO₂ lattice owing to the thermodynamic chemical potential driving force. Fig. 6 summarizes some typical examples of chemical bond formation energy between different metals and HfO₂ [118]. Owing to the oxygen scavenging effect, certain oxygen gradient are formed at interface between the active electrode and metal oxide. This interface layer [referred to as the oxygen exchange layer (OEL)] serves as an oxygen reservoir during SET/RESET operation. It stores oxygen after SET and supplies oxygen during RESET. In this sense, the electrode is part of the switching stack, rather than merely an electrical contact.

D. Asymmetry Stack for 1 and 0 States

Besides using active metals to form OEL, oxide-to-oxide stacks are another common way. Typical stacks are based on oxides with many stoichiometries, i.e., TaO_x/Ta₂O₅,

TABLE IV
TYPICAL OXRAM STACKS WITH ASYMMETRIC OXYGEN PROFILE
CREATED BY DIFFERENT PROCESSES

Asymmetric stacks	Materials	Deposition technique	Oxygen rich side	Vacancy side	rich SET/RESET polarity
Stoichiometry	HfO ₂ + Ti [13, 123]	ALD + PVD	HfO ₂	Ti	SET: + Ti Reset: + HfO ₂
Oxide + reactive metal cap	HfO ₂ + Zr [15]	ALD + PVD	HfO ₂	Zr	SET: + Zr Reset: + HfO ₂
	HfO ₂ + Hf [17, 124]	ALD + PVD	HfO ₂	Hf	SET: + Hf Reset: + HfO ₂
	Al ₂ O ₃ + TiO ₂ [125]	N/A	Al ₂ O ₃	TiO ₂	SET: + TiO ₂ Reset: + Al ₂ O ₃
Non-stoichiometry oxides with varying chemical composition	CuO _x [11, 55]	Thermal oxidation / plasma oxidation	CuO	Cu _x O	SET: + Cu _x O Reset: + CuO
	WO _x [49]	Oxidation	WO ₃	W ₂ O ₅ , WO ₂	SET: + W ₂ O ₅ , WO ₂ Reset: + WO ₃
	FeO _x [50]	RF sputtering + anneal	Fe ₂ O ₃	γ-Fe ₂ O ₃	SET: + γ-Fe ₂ O ₃ Reset: + Fe ₂ O ₃
	TaO _x [53]	RF sputtering + oxygen anneal	Ta ₂ O ₅	TaO ₂	SET: + TaO ₂ Reset: + Ta ₂ O ₅
	CoO _x [76]		Co ₃ O ₄	CoO	SET: + CoO Reset: + Co ₃ O ₄
	CeO _x [59]	Pulse laser deposition in oxygen	CeO ₂	Ce ₂ O ₃	SET: + Ce ₂ O ₃ Reset: + CeO ₂
Binary oxides + noble metal	HfO ₂ [89] ZrO ₂ [88]	ALD/PVD/thermal oxidation	Noble metal	TMO	SET: + oxides Reset: + noble metal

WO_x/WO₃, etc. This method is similar to tuning the metal-to-oxygen ratio during reactive sputtering. A general description on the bipolar switching OxRAM stack is that there is an asymmetry of oxygen contents within the stack. With the asymmetry of oxygen contents, an OEL layer is built in and serves as an oxygen reservoir. The asymmetric oxygen profile within the stack chemically creates two different domains, with chemical potential difference for oxygen to metal bonding. With this asymmetry, two different thermodynamic energy states are created. Resistive switching then occurs between these two states. Table IV summarizes some typical bipolar switching stacks with asymmetric material regions/compositions within the stack.

V. ReRAM BENCHMARK

A. Characteristics Benchmark

By summarizing the basic switching and reliability properties in International Electron Device Meeting (IEDM), very large scale integration (VLSI), and International Memory Workshop (IMW) publications between 2004 and 2019 [10]–[15], [17]–[22], [24], [30]–[32], [40], [42], [43], [45]–[79], [108], [117], [122], [127]–[150], a general trend of the ReRAM switching characteristics can be drawn.

B. Switching Current

Fig. 7 shows the SET versus RESET current trend for ReRAM devices. For typical bipolar filamentary switching, the maximum switching current level for the SET and RESET operations are symmetric. Most of the demonstrations show the switching current equal to or above 10 μ A. For stand-alone applications (NAND replacement or SCM), sub-10- μ A operating current is pursued. However, with the switching current scaling in filamentary switching, retention, nonuniformity, and endurance degrade severely. This is one of the reasons that most of the published demonstrations on filamentary switching are above 10 μ A. At sub-10 μ A, the symmetric trend of SET and RESET currents also breaks apart, indicating a different mechanism other than filamentary switching. Fig. 7 lists the low current switching demonstrations below 10 μ A, and most

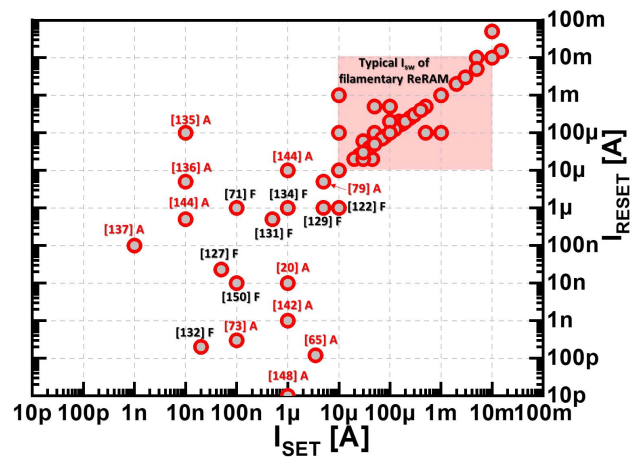


Fig. 7. SET and RESET switching current summary. Filamentary and area switching demonstrations below 10 μ A are marked with “F” and “A,” respectively.

of them are area switching cells rather than filamentary switching. With more self-rectify feature (nonlinear device $I-V$) built in these area switching cells, the SET and RESET currents are usually asymmetric.

C. Switching Speed

Thanks to the fast ionic movement in the nanometer thin film, SET/RESET switching demonstrated on single bit and mini array shows sub-100 ns characteristics. Though on the actual array level with megabit range bank size, the SET/RESET and READ latency are mostly dominant by the RC of the bitline (BL) and word line (WL), as well as by the periphery circuit. It is still fair to say that the ReRAM cell itself does not add up significant overhead to the total latency. The total SET/RESET latency also depends on the actual program scheme. READ verification after SET/RESET switching [i.e., incremental step pulse programming (ISPP)] is commonly used in nonvolatile memory (NVM) (i.e., NAND flash). And the total verified attempts define the actual SET/RESET latency. The READ verification attempts at the medium switching current (50 μ A) in TMO-based OxRAM stays in a low range ($\sim 5\times$) [139], whereas at the 10- μ A switching current, the number of attempts increases as the switching is more difficult to reach the target in one single shot. For high current operation (150–200 μ A), single SET/RESET pulses with very minor verify scheme shall deliver enough LRS/HRS window, which enables fast READ and fast SET/RESET on the array level.

D. Switching Voltage

The switching voltage on various ReRAM devices spreads out in a wide range. The actual switching voltage depends on ion mobility, material stack thickness, parasitic resistance, etc. Thus, it is not meaningful to benchmark the switching voltage value itself, but rather showing the general trend of voltage versus speed tradeoff. A general exponential dependence between voltage versus speed is seen in Fig. 8 for ReRAM switching [151].

The actual switching voltage value is an important parameter for ReRAM in embedded application. Typically, ReRAM is

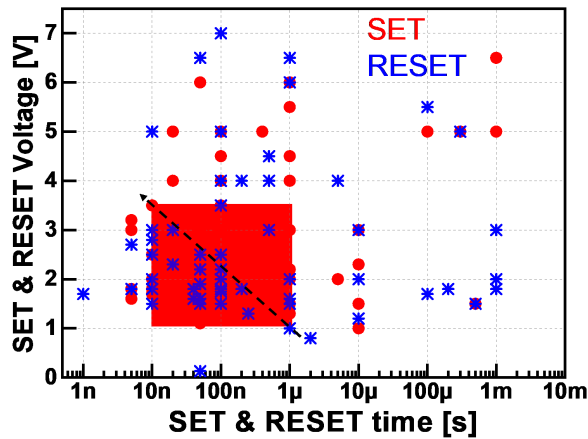


Fig. 8. SET/RESET voltage versus SET/RESET switching time summary. A general exponential dependence trend is observed.

driven by a transistor as BEOL NVM in embedded application (referred to as 1T1R). With the logic CMOS scaling, the V_{DD} of the MOSFET scales down (i.e., 40-nm foundry CORE 0.9–1.2 V, I/O 1.8–2.5 V [152]). The logic transistor voltage compatibility sets the practical limits for the ReRAM operation voltage, which requires cooptimization in 1T1R integration.

E. Endurance

Filamentary OxRAM has demonstrated up to 10^{12} cycles [18], [78], and the number of other demonstrations shows 10^{10} cycles [69], [77]. Generally, higher current with balanced SET/RESET conditions gives better endurance. Cautions have to be taken on the sampling READ approach typically used in the long endurance test. Owing to its defect-based nature, ReRAM may show random switching failure during cycling. Many of these failures are recoverable and the device can continue cycling. The READ verify approach is more accurate to assess the endurance lifetime.

For the low-end embedded application that ReRAM targets for, 10^4 cycle is typically the lower limit for its endurance lifetime. Since eNVM is able to have the current supply $>200 \mu\text{A}$ (higher current generally leads to better endurance, as shown in Fig. 9), the 10^4 cycle endurance spec is a relatively safe number for initial product introduction.

For NAND replacement, the endurance spec spreads between a few thousands and 10^6 cycles, depending on what types of NAND (TLC versus SLC) technology it is compared with. Though the hope is that ReRAM can reach more than 10^6 cycles, to offer better endurance with respect to NAND, for SCM application, 10^6 cycle is the typical spec. Note that for both NAND and SCM applications, the switching current spec is scaled to $10 \mu\text{A}$. At such low switching current, there are few reports of long endurance. This is one limitation, from device characteristic point of view, that blocks the way for ReRAM in the stand-alone application.

A few more in-depth studies investigate the endurance failure mechanism on OxRAM [154], [155]. Failure depends on the switching current level, SET/RESET conditions, and material stacks. A unified failure model is difficult to conclude. However, the general trend is that, for the high switching current, failing to RESET is one of the common modes. The dielectric layer is damaged during the SET/RESET operations

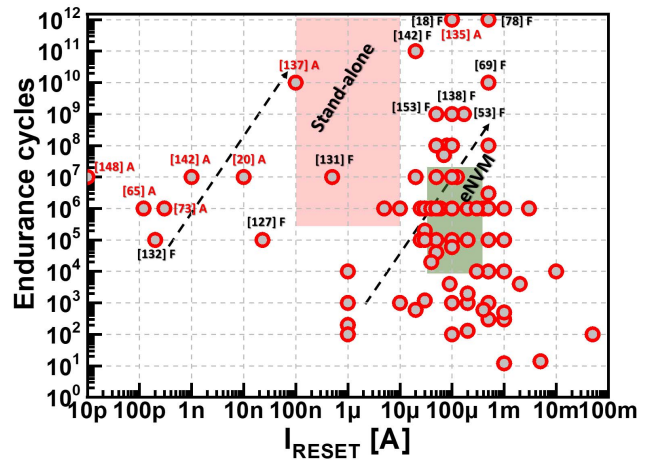


Fig. 9. Demonstrated ReRAM endurance versus RESET switching current summary. Filamentary and area switching demonstrations with current lower than $10 \mu\text{A}$ and endurance higher than 10^9 cycles are marked with “F” and “A,” respectively.

and gradually results in the permanent dielectric breakdown. For low and medium switching currents, failure modes vary. Both failing to RESET and failing to SET have been seen. With switching current reduction, the filament contains less defects and the failure behavior is more random.

F. Retention

Retention benchmark on ReRAM is not straightforward. Fundamentally all technologies have certain amount of bit failure at different temperatures. The definition and statistics of failure criteria are the two essential items of the retention benchmark. Since majority of the ReRAM research work only demonstrate single bit retention, the benchmark in Fig. 10 is for the reference purpose only. It is worth mentioning that Fig. 10(a) is plotted with the switching current on the X-axis, which is the determining factor in ReRAM retention.

From the array statistics viewpoint, from the kilobit to megabit range for the embedded application (presumably $\sim 200 \mu\text{A}$ range), ten-year-retention at 85°C [28], [143], [147] is a reasonably good bit error rate (BER) ($<10^{-7}$) [30] and even ten-year-retention at 125°C [146] [149] is achievable. The extracted activation energy of retention failure is in the range of 1.2 eV [Fig. 10(b)], which is similar to the FG NAND flash.

With switching current reduction, retention severely degrades [40] in filamentary switching ReRAM. According to Fig. 10, demonstration for long retention with the sub- $50\text{-}\mu\text{A}$ switching current is limited even on a single-bit level. For the stand-alone application with the targeted switching current below $10 \mu\text{A}$, the programmed LRS and HRS lose their resistance values even at room temperature within seconds [155]. The fast retention loss at low switching current is usually referred to as relaxation. Practically, there is no straightforward way to mitigate relaxation without tradeoff of SET/RESET latency. Relaxation is another major limitation for ReRAM in a stand-alone application.

Fig. 11 compares the retention and relaxation properties between OxRAM and CBRAM. Though both OxRAM and CBRAM show strong retention degradation at the $10\text{-}\mu\text{A}$

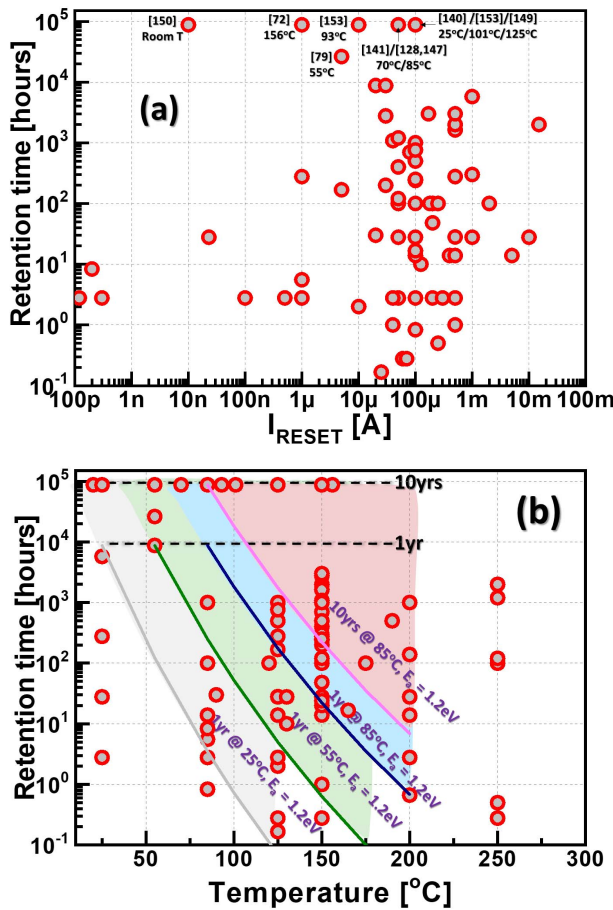


Fig. 10. (a) Demonstrated ReRAM retention versus RESET switching current summary. (b) Demonstrated ReRAM retention versus temperature summary. Calculated guideline with 1.2-eV activation energy for ten years at 85 °C, one year at 85 °C, one year at 55 °C and one year at 25 °C are plotted.

switching current, the degradation at the 25–50- μA range is less pronounced for CBRAM. It indicates the fact that the filament conduction with oxygen vacancies and that with metal ions are not exactly the same, in terms of defect density or defect conductivity. The other difference between OxRAM and CBRAM is that OxRAM filament FORMING is a controlled dielectric breakdown event, whereas metal ion injection into the dielectric in CBRAM FORMING can occur well before the applied voltage reaches the dielectric breakdown voltage. Thus, the metal ion filament formation in CBRAM can be decoupled from the dielectric breakdown. A less damaged dielectric with less oxygen vacancies results in less randomness and probably gives better stability at the medium current range (25–50 μA).

Owing to the limitation on retention at the scaled switching current, area-dependent switching devices have been studied as an alternative. Fig. 12 shows the retention comparison of the OxRAM and the area switching (a-VMCO [80]) device. The area-dependent switching ReRAM clearly outperforms the filamentary ReRAM in relaxation. With a higher number of defects involved in the switching, the LRS is better maintained. However, the endurance of area-dependent devices at scaled dimension (40-nm CD) with a verification scheme shows the limited number (1500 cycles) [158], which is far below the SCM spec.

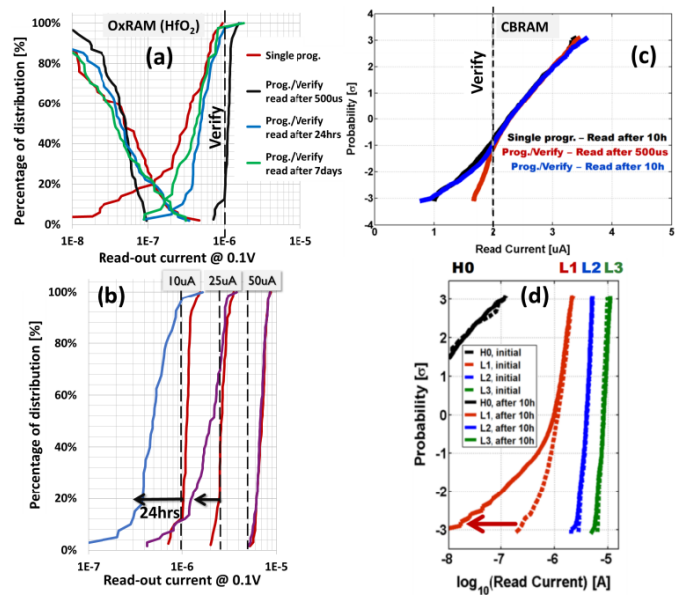


Fig. 11. Relaxation (fast retention loss at room temperature) phenomena comparison between (a) and (b) OxRAM [156] and (c) and (d) CBRAM [157]. LRS is programmed with program and verify to a predefined read current level (as indicated by the dashed line “Verify”) and read operations are carried out after different waiting time. The first read at 500 μs after LRS program already shows a fast retention loss, with LRS distribution relaxes to the level of single pulse program without verify (a) and (c). The relaxation shows strong current level dependence. Low program current level (10 μA) shows clear relaxation while medium current level (50 μA) shows stable retention (b) and (d).

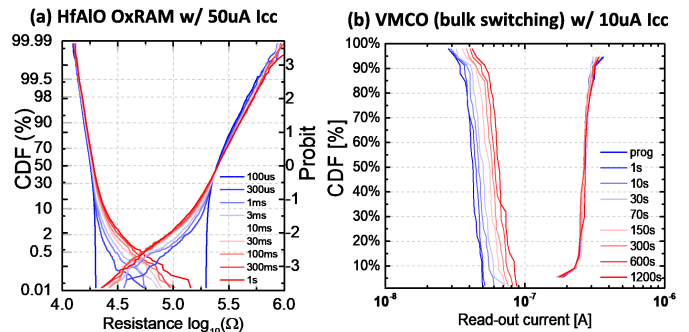


Fig. 12. (a) Relaxation of filamentary OxRAM [155]. (b) Relaxation of area switching ReRAM [80].

VI. COMMERCIALIZATION OF ReRAM

The revival of ReRAM from the early 2000s was clearly aimed at a storage technology with better performance and lifetime than the NAND flash. A 2-D FG NAND was scaling well, from 90 nm node in 2002 to 24 nm node in 2011 [9], totalling six generations within one decade. However, there was always a concern, which was if the 2-D NAND scaling was going to hit the limit.

Such concern was strong. *Device physics wise*, the number of electrons on the NAND gate was scaling with the device CD. With projection that when it reached 15-nm CD (1Z nm node for 2-D NAND), the electron static control on the multiple programmed V_T levels would see a serious overlap. There were only the single-digit numbers of electrons for each V_T level, and the electrical impact on adding or losing one electron became deterministic for the individually programmed

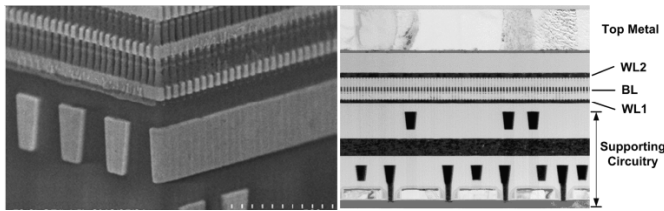


Fig. 13. SanDisk/Toshiba 2-deck 24-nm ReRAM demonstration [161], [162].

V_T level. The same challenge was amplified by interference from densely packed neighboring bits, which were getting closer and closer with the pitch scaling.

Process technology wise, 2-D NAND flash CD scaling was driven by lithography. 2-D NAND led the semiconductor industry to use 193-nm immersion lithography and self-aligned multiple patterning [self aligned double patterning (SADP), self aligned quadruple patterning (SAQP), etc.] for dimensional scaling. It was still possible for the same SAQP scheme to continue the lithography printing down to 10 nm, but the process control became more critical.

Architecture wise, 2-D NAND relied on the Si substrate as a channel material. Stacked 2-D NAND with either the c-Si channel [159] or poly-Si thin-film transistor (TFT) was demonstrated [160], but potentially had thermal budget constraint for the multiple-layer process and higher process cost.

With the consideration to extend the storage roadmap when NAND scaling was running out of steam, ReRAM as a pure $4F^2$ memory cell, with potential to be stacked in a simple cross-point array (BL and WL perpendicular to each other), drew attentions. The filamentary nature made the switching itself almost device CD independent, which was perfect for scaling further. Material stacks were usually thinner than 10 nm, making etching easier at the scaled pitch. The switching voltage was much lower than the NAND flash and single-bit switching speed was much faster. More importantly, the simple cross-point array made individual ReRAM bits randomly accessible. A true RAM with nonvolatility at scaled dimension could be enabled. All these properties made ReRAM a good candidate to replace the NAND flash.

A. Stand-Alone

1) NAND Replacement: Following the NAND replacement discussion, there were a number of efforts within the semiconductor memory industry in pursuing ReRAM as high-density stand-alone memory in a stackable architecture. Fig. 13, reprinted from the SanDisk/Toshiba demonstration [161], [162], shows a two-deck ReRAM at a 24-nm node. The stacked cross-point memory architecture was first commercialized by a Matrix semiconductor with their one time programmable (OTP)-based ROM-type memory [170]. With its acquisition by SanDisk in 2005, the stacked cross-point architecture was then coupled with ReRAM. For NAND replacement, the initial anticipated insertion point was equivalent to 43-nm 2-D NAND (commercialization in 2008). The 43-nm 2-D NAND is a 16 Gb (one plane)/32 Gb (two planes) density with TLC technology. From the density point of view, the introduction of TLC NAND raised the market entry bar of ReRAM technology. To compete with MLC NAND, a two-deck ReRAM at the same node could have more scaled die size with a smaller periphery. To compete with TLC NAND,

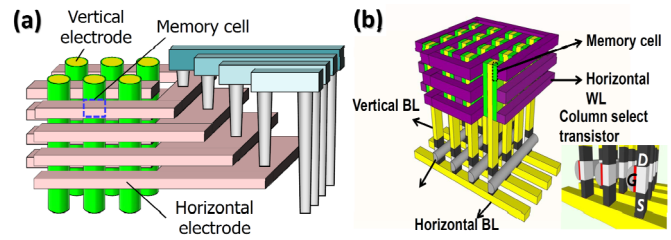


Fig. 14. 3-D vertical ReRAM architecture proposals. (a) [128]. (b) [166]. BLs are vertically oriented, and WLs are horizontally extended as comb shape.

potentially a four-deck ReRAM was needed. The continuous logic state scaling in NAND technology, from SLC to MLC to TLC, kept the requirement of ReRAM higher and higher. Together with the switching current versus retention/endurance issues discussed in Section V, 2-D NAND replacement became unachievable.

On the other hand, NAND scaling continues. The introduction of fully vertical 3-D NAND technology (referred to as BiCS) by Toshiba in 2007 [26] successfully extends the NAND scaling roadmap after 1Z nm 2-D NAND. Currently, the industry is shipping $9\times$ layers 3-D NAND with QLC technology. Single die density reaches 1.33 Tb [163]. The 2-D memory stacking concept at 20-nm CD probably needs more than 20 decks to reach such density.

2) SCM: With the NAND replacement requirement gradually faded away, ReRAM saw another potential application as a new memory layer within the memory hierarchy to bridge the latency gap between DRAM and NAND: SCM. The concept was proposed for some years already [164], while no technology came close to its requirement [165]. 10^6 cycle endurance with certain nonvolatility (months to a year) and density/latency between DRAM and NAND are needed for SCM.

With the density requirement and the fact that most of the ReRAM devices could only operate reliably in SLC, multiple innovation on a 3-D vertical ReRAM technology was proposed [14], [128], [142], [166], [173], [174] (Fig. 14). They all have similar vertical architectures, with vertical BL and horizontal WL in a flat trench type of structure. It is similar to some alternative 3-D NAND architectures [167], [168], which were not adopted. With the architecture innovation, the device requirement, however, became stricter. The device shall have certain nonlinearity to enable a selector-less array operation so that the device stack was thin enough to be placed inside the vertical BL trench/hole. The nonlinearity requirement increased with a higher number of WL layers. At the same time, the switching current shall maintain at $10\ \mu\text{A}$ or below for power and leakage spec. The READ current shall be as high as possible to enable low READ latency for SCM. In a two-terminal device, such as ReRAM, the switching and READ paths were always coupled. Thus, the low switching current and high READ current requirements contradicted each other. In case the memory cell did not provide enough nonlinearity, a separate selector was needed. Memory to selector cooperation at low current switching and vertical process integration triggered more difficulties. Since the switching current requirement at the sub- $10\text{-}\mu\text{A}$ level still held for ReRAM as SCM application, the fundamental LRS/HRS relaxation and retention issues remained unsolved.

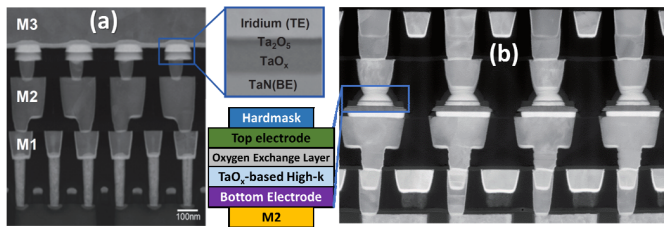


Fig. 15. (a) [24] ReRAM 1T1R cell embedded between M2/M3 on 40-nm CMOS logic (ReRAM physical size 117 nm). (b) [30] ReRAM 1T1R embedded between M2 / M4 on 22-nm FinFET CMOS logic (ReRAM physical size 100 nm).

Owing to all this, ReRAM faces practical challenges to be used for SCM applications.

B. eNVM on Logic

Following many of the early demonstrations of ReRAM in the 1T1R configuration, these technologies are naturally extended to eNVM integrated at logic BEOL. Panasonic has been pioneering their TaO_x-based ReRAM in micro controller units (MCUs), smart cards, and other types of applications as NOR flash replacement [24]. TSMC has ReRAM as low-end eNVM for NOR flash replacement on their 40-nm logic node [28]. Intel demonstrated their ReRAM as eNVM on the 22-nm FinFET node [30]. Winbond has their HfO₂-based ReRAM on 90-nm CMOS technology as NVM offering [143] (Fig. 15). With respect to embedded NOR flash, ReRAM has clear cost advantage [28]. The retention spec may not be sufficient for automobile products, but good enough for most of the consumer products. Thanks to its lower energy consumption and lower cost, IoT devices deployed in the field with energy supply constraint can benefit from it.

Though the eNVM market is much smaller than stand-alone NAND, and the largest segment within eNVM is automobile products which ReRAM is still not able to compete with NOR flash in reliability, the commercialization of ReRAM in this fully established market proves its technology advantage. Simple process, low cost, and sufficient reliability at high current operation are proven by the market. With ReRAM potentially to be offered on more advanced logic nodes (i.e., 28-nm high k metal gate (HKMG), 16-/14-nm FinFET, etc.), the market adoption can be further expanded.

C. Other Attempts

Besides the efforts from established industry players, a number of start-ups have been also working on different flavors of ReRAM technology for commercialization. Unity (acquired by Rambus in 2012) has demonstrated SRC switching ReRAM at 64-Mb density with 0.13- μ m technology [170]. Adesto [21], [130], has been continuously working on CBRAM technologies with different material stacks. Cross-bar [171] tries to combine ReRAM with their high ON/OFF ratio threshold selector technology. 4DS [172] is working on their area switching ReRAM technology for SCM application.

Another player once in the field to be noted is HP [84]. With their work on the memristor, as using ReRAM in nature, but applying as the circuit element rather than digital memory, ReRAM becomes more popular beyond the traditional memory technology community.

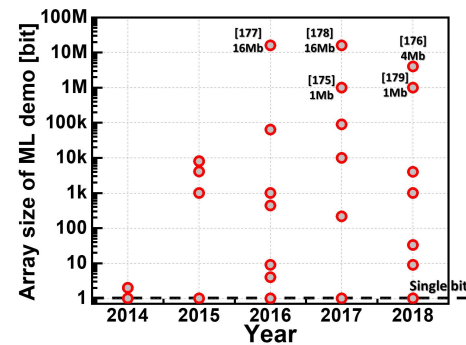


Fig. 16. ReRAM array demonstration for machine learning application in IEDM and VLSI between 2014 and 2018, with [175]–[179] highlighted.

VII. ReRAM FOR MACHINE LEARNING

With the wave down of ReRAM research as memory technology over the past five years, more attempts are made on exploring the potential of using it for machine learning applications. From early attempts using ReRAM as a digital binary switch element [177] to “compute in memory, CiM” using ReRAM as both storage and computing element, the motivation always stays in leveraging its simple and high-density cross-point architecture. Fig. 16 summarizes the ReRAM technology demonstration with different machine learning implementations and algorithms. With technology point of view, encouraging progress has been made from single-bit engineering work with array simulation to kilobit and megabit array-level demonstrations. With challenges ahead to scale up the implementation to even large bit counts, more efforts are yet to be made.

VIII. CONCLUSION

Significant efforts have been made over the past 20 years in the field of ReRAM technology research and development, with aims to commercialize it and understand it. As of today, its adoption is still limited, and its understanding is still incomplete. ReRAM technology offers many unique properties worth the research and development efforts, as well as helps overcome difficult scaling barriers. However, with the recent interest in energy-efficient machine learning applications, we may once again see a rise in ReRAM sentiment.

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